Exercises and Problems for Section 8.3

Exercises

Convert the polar coordinates in Exercises 1–4 to Cartesian coordinates. Give exact answers.

1. $(1, 2\pi/3)$

2. $(\sqrt{3}, -3\pi/4)$

3. $(2\sqrt{3}, -\pi/6)$

4. $(2, 5\pi/6)$

Convert the Cartesian coordinates in Exercises 5–8 to polar coordinates.

5. $(1, 1)$

6. $(-1, 0)$

7. $(\sqrt{6}, -\sqrt{2})$

8. $(-\sqrt{3}, 1)$

9. **(a)** Make a table of values for the equation $r = 1 - \sin\theta$. Include $\theta = 0, \pi/3, \pi/2, 2\pi/3, \pi, \cdots$.
(b) Use the table to graph the equation $r = 1 - \sin\theta$ in the xy-plane. This curve is called a *cardioid*.
(c) At what point(s) does the cardioid $r = 1 - \sin\theta$ intersect a circle of radius 1/2 centered at the origin?
(d) Graph the curve $r = 1 - \sin 2\theta$ in the xy-plane. Compare this graph to the cardioid $r = 1 - \sin\theta$.

10. Graph the equation $r = 1 - \sin(n\theta)$, for $n = 1, 2, 3, 4$. What is the relationship between the value of n and the shape of the graph?

11. Graph the equation $r = 1 - \sin\theta$, with $0 \le \theta \le n\pi$, for $n = 2, 3, 4$. What is the relationship between the value of n and the shape of the graph?

12. Graph the equation $r = 1 - n\sin\theta$, for $n = 2, 3, 4$. What is the relationship between the value of n and the shape of the graph?

13. Graph the equation $r = 1 - \cos\theta$. Describe its relationship to $r = 1 - \sin\theta$.

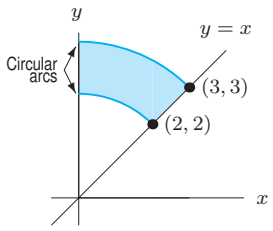
14. Give inequalities that describe the flat surface of a washer that is one inch in diameter and has an inner hole with a diameter of 3/8 inch.

15. Graph the equation $r = 1 - \sin(2\theta)$ for $0 \le \theta \le 2\pi$. There are two loops. For each loop, give a restriction on θ that shows all of that loop and none of the other loop.

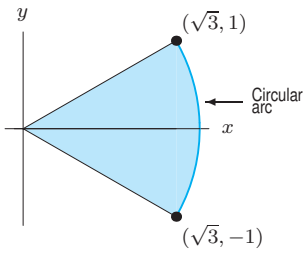
16. A slice of pizza is one eighth of a circle of radius 1 foot. The slice is in the first quadrant, with one edge along the x-axis, and the center of the pizza at the origin. Give inequalities describing this region using:
(a) Polar coordinates **(b)** Rectangular coordinates

In Exercises 17–19, give inequalities for r and θ which describe the following regions in polar coordinates.

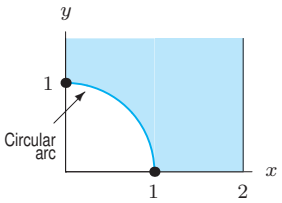
17.



18.



19.



Note: Region extends indefinitely in the y-direction.

20. Find the slope of the curve $r = 2$ at $\theta = \pi/4$.

21. Find the slope of the curve $r = e^{\theta}$ at $\theta = \pi/2$.

22. Find the slope of the curve $r = 1 - \cos\theta$ at $\theta = \pi/2$.

23. Find the arc length of the curve $r = e^{\theta}$ from $\theta = \pi/2$ to $\theta = \pi$.

24. Find the arc length of the curve $r = \theta^2$ from $\theta = 0$ to $\theta = 2\pi$.

Problems

25. Sketch the polar region described by the following integral expression for area:

$$\frac{1}{2}\int_0^{\pi/3} \sin^2(3\theta)\, d\theta.$$

26. Find the area inside the spiral $r = \theta$ for $0 \le \theta \le 2\pi$.

27. Find the area between the two spirals $r = \theta$ and $r = 2\theta$ for $0 \le \theta \le 2\pi$.

28. Find the area inside the cardioid $r = 1 + \cos\theta$ for $0 \le \theta \le 2\pi$.

29. **(a)** In polar coordinates, write equations for the line $x = 1$ and the circle of radius 2 centered at the origin.
(b) Write an integral in polar coordinates representing the area of the region to the right of $x = 1$ and inside the circle.
(c) Evaluate the integral.

30. Show that the area formula for polar coordinates gives the expected answer for the area of the circle $r = a$ for $0 \le \theta \le 2\pi$.

31. Show that the arc length formula for polar coordinates gives the expected answer for the circumference of the circle $r = a$ for $0 \le \theta \le 2\pi$.

32. Find the area inside the circle $r = 1$ and outside the cardioid $r = 1 + \sin\theta$.

33. Find the area inside the cardioid $r = 1 - \sin\theta$ and outside the circle $r = 1/2$.

34. Find the area lying outside $r = 2\cos\theta$ and inside $r = 1 + \cos\theta$.

35. **(a)** Graph $r = 2\cos\theta$ and $r = 2\sin\theta$ on the same axes.
(b) Using polar coordinates, find the area of the region shared by both curves.

36. For what value of a is the area enclosed by $r = \theta$, $\theta = 0$, and $\theta = a$ equal to 1?

37. **(a)** Sketch the bounded region inside the lemniscate $r^2 = 4\cos 2\theta$ and outside the circle $r = \sqrt{2}$.
(b) Compute the area of the region described in part (a).

38. Using Example 11 on page 437, find the equation of the tangent line to the curve $r = 3\sin 2\theta$ at $\theta = \pi/3$.

39. Using Example 11 on page 437 and Figure 8.43, find the points where the curve $r = 3\sin 2\theta$ has horizontal and vertical tangents.

40. For what values of θ on the polar curve $r = \theta$, with $0 \le \theta \le 2\pi$, are the tangent lines horizontal? Vertical?

41. **(a)** In Cartesian coordinates, write an equation for the tangent line to $r = 1/\theta$ at $\theta = \pi/2$.
(b) The graph of $r = 1/\theta$ has a horizontal asymptote as θ approaches 0. Find the equation of this asymptote.

42. Find the maximum value of the y-coordinate of points on the limaçon $r = 1 + 2\cos\theta$.

Find the arc length of the curves in Problems 43–44.

43. $r = \theta$, $0 \le \theta \le 2\pi$

44. $r = 1/\theta$, $\pi \le \theta \le 2\pi$

45. For the curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$, show that

$$\text{Arc length} = \int_\alpha^\beta \sqrt{(f'(\theta))^2 + (f(\theta))^2}\, d\theta.$$

46. Find the arc length of the spiral $r = \theta$ where $0 \le \theta \le \pi$.

47. Find the arc length of part of the cardioid $r = 1 + \cos\theta$ where $0 \le \theta \le \pi/2$.

Strengthen Your Understanding

In Problems 48–51, explain what is wrong with the statement.

48. The point with Cartesian coordinates (x, y) has polar coordinates $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}(y/x)$.

49. All points of the curve $r = \sin(2\theta)$ for $\pi/2 < \theta < \pi$ are in quadrant II.

50. If the slope of the curve $r = f(\theta)$ is positive, then $dr/d\theta$ is positive.

51. Any polar curve that is symmetric about both the x and y axes must be a circle, centered at the origin.

In Problems 52–55, give an example of:

52. Two different pairs of polar coordinates (r, θ) that correspond to the same point in the plane.

53. The equation of a circle in polar coordinates.

54. A polar curve $r = f(\theta)$ that is symmetric about neither the x-axis nor the y-axis.

55. A polar curve $r = f(\theta)$ other than a circle that is symmetric about the x-axis.

8.4 DENSITY AND CENTER OF MASS

Density and How to Slice a Region

The examples in this section involve the idea of *density*. For example,

- A population density is measured in, say, people per mile (along the edge of a road), or people per unit area (in a city), or bacteria per cubic centimeter (in a test tube).
- The density of a substance (e.g. air, wood, or metal) is the mass of a unit volume of the substance and is measured in, say, grams per cubic centimeter.

Suppose we want to calculate the total mass or total population, but the density is not constant over a region.